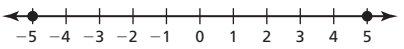
Answers

Chapter 8

8.1 Lesson Extra Practice

1. -5 **2.** 7
3. -35 **4.** -50
5. -12 **6.** 10 **7.** 15
8.


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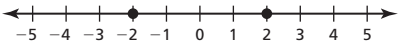

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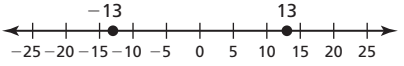

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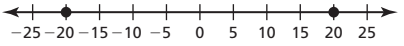

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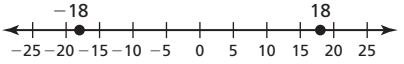

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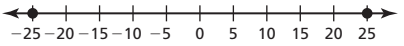
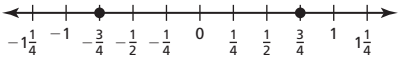


14. no; The opposite of 7 is -7.
15. -10 **16.** 17
17. 8 **18.** -17
19. **a.** 2; yes **b.** -5; no **c.** -1; no
20. **a.** the elevation at the start of the trail
b. -4

8.2 Lesson Extra Practice

1. $<$ **2.** $>$
3. $>$ **4.** $<$
5. $<$ **6.** $<$
7. no; $2 > -5$ **8.** $-3, -2, 0, 2, 3$
9. $-4, -3, 1, 2, 4$ **10.** $-5, -4, 3, 4, 5$
11. $-5, -3, 1, 4, 6$
12. $-3 > -8$, so the water pipe is above the gas pipe.
13. 2 **14.** never
15. never **16.** sometimes
17. Throughout winter, spring, and summer, the amount of rainfall was higher than the rainfall of the previous year during those seasons. During the fall, the amount of rainfall was lower than the rainfall in the fall of the previous year.
18. your friend
19. **a.** $-16, -13, -6, -5, 1, 6, 12$
b. The other two integers are -6 or less.

8.3 Lesson Extra Practice

1.


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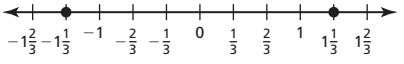

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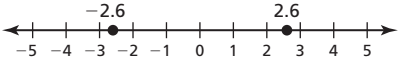

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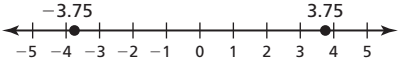

5. $-\frac{10}{3} < -\frac{7}{4}$ **6.** $\frac{4}{5} > -1\frac{7}{8}$
7. $-\frac{7}{6} < -\frac{6}{7}$ **8.** $-2\frac{3}{4} < -2\frac{2}{3}$
9. $2.1 > -2.1$ **10.** $-0.08 > -0.8$
11. $-3.08 > -4.16$ **12.** $-4.82 > -4.89$
13. $-\frac{7}{10}, -\frac{1}{2}, -\frac{2}{5}, -\frac{3}{10}, -\frac{1}{10}$
14. $-\frac{3}{12}, -\frac{2}{12}, \frac{7}{12}, 1\frac{1}{12}, 1\frac{5}{12}$
15. $-2\frac{2}{3}, -2, -1\frac{2}{3}, -1\frac{1}{3}, -\frac{2}{3}$
16. $-3, -2.75, -2.4, -2.1, -2$
17. first
18. -2.83 km, -2.70 km, -2.48 km, -2.15 km
19. **a.** 2021–2022
b. 2017–2018

8.4 Lesson Extra Practice

1. 13 **2.** 8
3. 4 **4.** 1
5. 5.2 **6.** 12
7. $2\frac{1}{3}$ **8.** 51
9. $\frac{5}{6}$ **10.** 40
11. no; $|-20| = 20$
12. $>$ **13.** $=$ **14.** $<$ **15.** $>$
16. **a.** your boat: 18
friend's boat 23
b. friend's boat
17. **a.** minute 4; minute 1
b. lower; 11.75 feet
18. $-2, 0, 1, |-3|, |5|$ **19.** $-5, -3, |-1|, |3|, |-5|$
20. always; When a is negative, $|a| = -a$.
21. never; $|a| \geq a$
22. never; Absolute value is never negative.

8.5 Lesson Extra Practice

1. $A(2, 3)$ **2.** $B(0, -4)$
3. $C(1, -3)$ **4.** $D(4, 1)$
5. $E(1, 0)$ **6.** $F(-3, -2)$
7. $G(-4, 3)$ **8.** $H(2, -4)$
9. $I(-2, -4)$ **10.** $J(-1, 1)$

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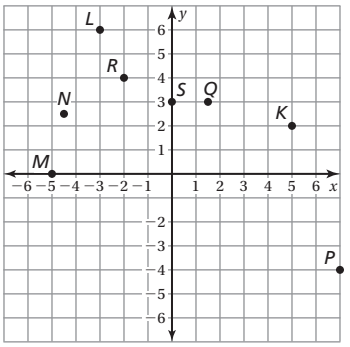
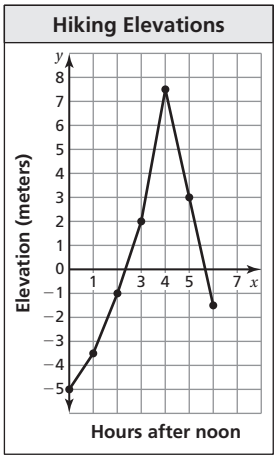
11–18.



11. Quadrant I **12.** Quadrant II
13. x-axis **14.** Quadrant II
15. Quadrant IV **16.** Quadrant I
17. Quadrant II **18.** y-axis
19. yes; The description is correct.
20. **a.** $(7, 1), (7, -1)$ **b.** $(-1, 3.5), (1, 3.5)$
21. **a.** $(0, 2), (0, -2)$ **b.** $(6, 4), (-6, 4)$
22. **a.** $(4, -1)$ **b.** $(-4, 1)$
23. **a.** $(-3, 2)$ **b.** $(3, -2)$
24. **a.** $(-5, 0)$ **b.** $(5, 0)$
25. **a.** $(2, -1)$ **b.** $(-4, 3)$ **c.** school
d. walk 6 blocks west and 4 blocks north
e. yes; No matter in which direction you move 2 blocks from the mall, you are still in Quadrant II.

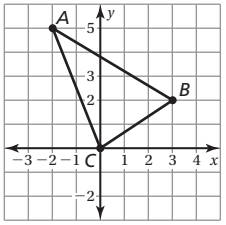
26.



The elevation increased from noon until 4:00 P.M. and decreased from 4:00 P.M. until 6:00 P.M.

8.6 Lesson Extra Practice

1.



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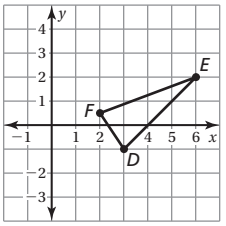

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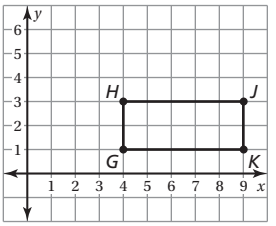

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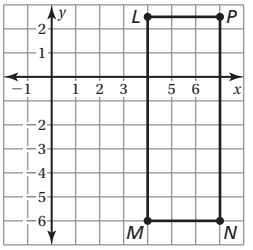
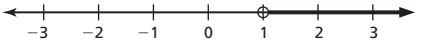


5. 5 **6.** 4
7. yes; The points are in different quadrants, so your friend adds the absolute values of the coordinates that are different.
8. 18 units; 14 units^2 **9.** 52 units; 168 units^2
10. **a.** $(5, 0)$ **b.** 24 yd, 32 yd^2
11. *Sample answer:* $D(10, 2)$; $E(6, 10)$

8.7 Lesson Extra Practice

1. $2 > v$ **2.** $h \le 15$
3. $p < \frac{1}{2}$ **4.** $12 < n$
5. $\frac{3}{5} \ge a$ **6.** $b \le 17$
7. $6 + x \ge 12$ **8.** $k - 7 > 10$
9. solution **10.** not a solution
11. not a solution **12.** not a solution
13. solution **14.** solution
15.



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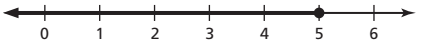

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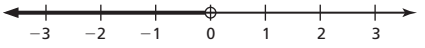

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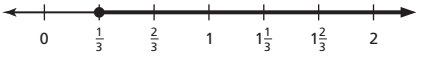

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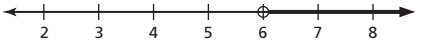

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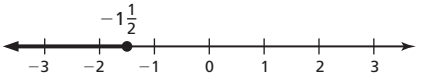

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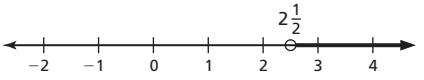

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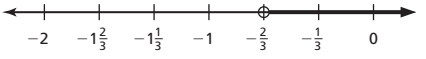
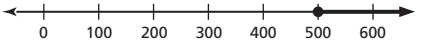
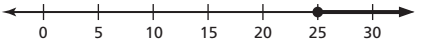


23.

24. no; 9 is not less than 9
25. $t \ge 500$;



26. **a.** $16t \le 50$
b. no; *Sample answer:* 4 is not a solution of $16t \le 50$.
27. **a.** Let w represent the number of pounds of potatoes; $w \ge 25$



b. yes

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